ZEYU CHEN

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SKILLS

Languages: Python, C++, C, JavaScript, Java, SQL, HTML, CSS, Solidity, WebAssembly, Bash
Frameworks & Libraries: OpenCV, NumPy, SciPy, Pandas, Tensorflow / Keras, PIL, Django, Jinja, Apollo GraphQL
Tools & Technologies: Git, Google Cloud Platform (GCP), Google Colab, Datadog, Jupyter Notebook, SageMath

EXPERIENCE

Coinbase :: Software Engineer Intern (Security Team) — **May 2022 - September 2022**

— Identified procedure deficiencies by conducting control gap analysis on InfoSec policies against **ISO 27001**
— Debugged and identified **HTML** iframe vulnerability in Chrome extension on transaction operations
— Disclosed sanitation rule for search bar to reduce SQL wildcard DoS attacks during penetration testing
— Created comprehensive documentation and schemas on policy dependencies for auditing and regulatory use
— Implemented **Ethereum** smart contract transactions using **Solidity**, authored as part of company-hosted CTF

DMOJ: Modern Online Judge :: Contest Organizer and Moderator — **July 2020 - Present**

— Co-authored **7** programming contests averaging **200+** participants each on Canada's largest platform
— Devised **graph theory**, **data structures** and **dynamic programming** problems with published solution editorials

PROJECTS

Colonel By Online Judge Programming Platform (cboj.ca) — **July 2020 - Present**

— Extended and open-sourced a platform hosted with **Google Cloud Platform (GCP)** for a high school computer science curriculum's assessments using **Python**, **Django**, **Jinja**, **HTML** and **CSS** — reaching **90+ daily users** in 2021
— Redesigned additional privacy levels with custom point-tallying using **Python** and **MySQL** for student assessments
— Monitored server with **Bash** to meet teachers' specifications, achieving **200+ days** of continuous uptime

3D Chess — **May 2021 - June 2021**

— Designed an autonomous player using **Minimax** search with alpha-beta pruning, **JavaScript**, **WebAssembly** and **C++**
— Reduced the search runtime by **over 70 seconds** per turn using pruning methods, heuristics and optimizations

Mole-croscope (CuHacking 2nd Place, Innovapost Challenge Winner) — **Feb 2019**

— Collaborated with a team of 4 to build a mole-tracking **Android** app with skin cancer prediction trained on **GCP**
— Designed mole anomaly-tracking through object translation invariance with **Python** and **OpenCV**
— Implemented **Python** preprocessing scripts to auto-correct skew and translation differences between user inputs

AWARDS

— Canadian Computing Olympiad (CCO 2021) :: **Silver medalist**
— Canadian Computing Competition (CCC 2021) :: **Perfect score, 1st place, top 1.1%** (~3 000 participants)
— Facebook Hacker Cup :: **Top 1.2%** in 2020, **top 1.6%** in 2021 (~35 000 participants)
— Google Kickstart Competition 2021 :: Round C **top 0.4%**, 48th worldwide (~12 200 participants)
— Google Kickstart Competition 2020 :: Round G **top 0.9%**, 73rd worldwide (~8 100 participants)
— Advent of Code :: **Top 0.2%** in 2019, **top 0.03%** in 2020, **top 0.017%** in 2021 (~210 000 participants)
— USA Computing Olympiad 2019, 2020, 2021 :: **Platinum division**

EDUCATION

University of Waterloo :: Bachelor of Software Engineering — **Sept 2021 - Present**

92.05% Cumulative Average, 3.93 / 4.0 GPA
Fall 2021 Term Distinction, Winter 2022 Term Distinction